

Question	Marking Guidance	Mark	Comments
05.1	1. Not caused by one gene; 2. (Hardy-Weinberg) cannot predict (the effect of) environmental factors OR Environmental factors involved; 3. (Various genetic factors suggests) mutations occur; 4. Human populations migrate/emigrate/immigrate;	2 max (2 x AO2)	Ignore any references to random mating 1. Ignore ideas relating to not knowing about dominant/recessive 2. Accept 'methylation/epigenetic factors' for 'environmental factors' 4. Accept descriptions of human migration 4. Accept human populations are not isolated 4. Ignore population size changes due to births/deaths

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05.2	1. Reduce/prevent expression/transcription (of <i>CDH1</i> /the gene); 2. Prevent transcription factor binding OR Prevent binding to promotor (of gene); 3. Reduce/prevent production of CDH1/protein;	1 max (1 x AO1)	Reject (methylation) causes mutation 1. Accept reduce/prevent production of messenger RNA/mRNA 1. Ignore 'switching off' 2. Accept prevent binding/activation of <u>RNA</u> polymerase

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05.3	1. Inflammation (alone) may be the cause as both (types of) treated cells have increased methylation; 2. Loss of <i>CDH1</i>/gene function might not have an effect as (both) untreated cells have similar (percentage) methylation; 3. Drug used to cause inflammation, so might not be the same as infection/diabetes; 4. Cells grown <i>in vitro</i>, so does not represent a living organism/human/body/fetus OR Cells grown <i>in vitro</i>, so might not respond in the same way as a living organism/human/body/fetus; 5. Cells not (allowed to) develop into lip/mouth/baby, so do not know any would develop CLP OR Cells not (allowed to) develop into lip/mouth/baby, so do not know what percentage (<i>CDH1</i>) methylation leads to CLP;	3 max (3 x AO3)	2. Accept loss of <i>CDH1</i>/gene function might not have an effect as 36% and 40% are similar 3. Accept 'normal environmental conditions' for 'infection/diabetes' 3. Accept 'artificial conditions' for 'drug' 4. Accept '<i>in vivo</i>' OR '<i>in utero</i>' OR 'pregnancy' for 'living organism' 4. Accept only cells grown, not a whole organism/human/body/fetus, so might not respond in the same way
